

Joseph E. Velasquez
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Education

California State University Fullerton

B.S. Computer Science

Riverside City College

A.S. Computer Science

Fullerton, CA

Aug 2025

Riverside, CA

May 2022

Experience

Inland Neurosurgery Institute - Software and Systems Engineer

Aug 2025 – Oct 2025

- Supported integration and deployment of a new Electronic Medical Records (EMR) system, ensuring minimal disruption to clinical operations
- Resolved hardware, software, and network issues during EMR migration to ensure continuity of clinical operations
- Implemented software updates to improve system performance and reliability
- Built and maintained automation workflows using Microsoft Power Automate, reducing manual administrative workload and increasing efficiency
- Collaborated with staff to enhance existing automation systems, streamlining processes and eliminating repetitive tasks
- Assisted in maintaining cybersecurity measures, including system monitoring, patching, and basic threat mitigation
- Developed a patient scheduling chatbot using local LLM, enabling automated appointment booking and reducing administrative workload

Residential IT Support Technician (ResNet) - University of California, Santa Barbara (UCSB)

Aug 2018 – June 2019

- Provided Tier 1 technical support for hardware, software, and network connectivity issues in a residential campus environment.
- Diagnosed and resolved Windows, macOS, Wi-Fi, Ethernet, TCP/IP, DNS, and DHCP issues.
- Configured laptops, smartphones, printers, gaming consoles, and other network-connected devices.
- Documented support requests and collaborated with team members to resolve issues efficiently.
- Refilled printers with paper across campus.

Projects

ParkingPulse - AI Vehicle Tracking & Data System

Aug 2024 – May 2025

- Engineered a real-time AI vehicle tracking platform using Python, OpenCV, and YOLOv8 for intelligent object detection and analytics.
- Designed and implemented RESTful APIs to handle real-time data ingestion, processing, and retrieval
- Integrated Firebase for authentication and cloud-based data storage, enabling scalable and secure data management
- Optimized system performance by reducing processing latency through frame skipping and model tuning
- Designed backend services capable of handling high-frequency event ingestion and real-time analytics workloads.
- Deployed cloud-hosted application components and monitored system performance for reliability and scalability.
- Implemented data pipelines for continuous processing and storage of vehicle detection events

Undergraduate Research Assistant — UAV Data & Vision Systems

Aug 2024 – Mar 2025

- Developed OpenCV-based computer vision pipelines for real-time object tracking in UAV simulations
- Processed geospatial routing data using OSMnx to improve autonomous navigation efficiency
- Simulated autonomous systems in Webots to evaluate navigation performance and system reliability
- Cleaned and structured large-scale datasets for computer vision and route optimization analysis
- Collaborated with research teams to improve tracking accuracy and system performance

Skills & Interests

- Languages: Python, JavaScript, TypeScript, C++, SQL
- Backend: Node.js, Flask, REST APIs
- Frontend: SvelteKit, HTML/CSS
- Cloud & DevOps: AWS (EC2, S3), Google Cloud Platform, Docker, CI/CD
- Databases: PostgreSQL, MongoDB, Firebase Firestore
- AI / Computer Vision: OpenCV, YOLOv8, Real-Time Data Processing
- Tools: Git, Linux

Certifications

- AWS Knowledge: Serverless
- Google Cloud: App Dev Environment
- Google Cloud: Load Balancing